Investigating the Transcriptome of Adenomatous Colorectal Polyps in the Context of Metachronous Recurrence

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on behalf of the Integrated Technologies for Improved Polyp Surveillance (INCISE) collaborators

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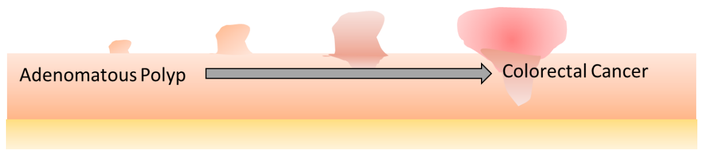
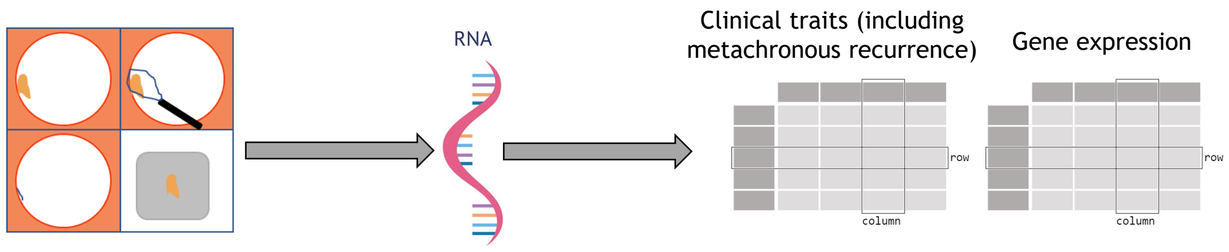
Background

Adenomatous colorectal polyps are composed of unhealthy, pre-cancerous cells, which display atypical genotypic and phenotypic characteristics derived from the cumulative incorporation of somatic mutations. In time, polyps may progress to colorectal cancer, thus they are strong prognostic indicators of disease risk.



Polypectomy, the surgical removal of diseased tissue, is the primary clinical intervention for adenomatous colorectal polyps and presents a strong opportunity for transcriptomic analysis. Gene signatures may be a method of patient stratification to identify those of risk of developing future polyps.

In the present retrospective study, we obtained both the transcriptome of resected index polyps and right censored information on metachronous of polyps over the following time-course (years).



We hypothesised that the transcriptomes of index polyps in individuals who develop downstream metachronous polyps would differ from non-metachronous individuals, and that these differences may be explained by semantic information on gene pathways and networks.

Aims:

1) Use differential expression analysis between metachronous and non-metachronous individuals to identify potential driver genes of polyp recurrence
2) Perform pathway and network analysis of these genes to infer biological processes
3) Use the ESTIMATE algorithm to examine the differences in stromal and immune cell infiltrates to these polyps
4) Use weighted-gene-co-expression-network-analysis (WGCNA) to identify co-expression networks which associate with, and may drive metachronous recurrence

Methods

Data Origin

All data was derived in-full as part of the Integrated Technologies for Improved Polyp Surveillance (INCISE) project, that aims to improve on current bowel cancer screening program in the UK. INCISE is primarily managed by the University of Glasgow [1].

Tissue Processing and Sequencing

- Surgical resection, largest polyp formaldehyde-fixed-paraffin-embedded immediately (FFPE).
- Sections from FFPE blocks submitted for probe based Tempo-Seq RNA Sequencing (mRNA only)
- Raw Tempo-Seq counts generated and concatenated for Genes x Patients

Stratification by Polyp Recurrence

A total of 1825 participants were randomly assigned to either a training (n=1379) or validation cohort (n=446). Count information for each patient was either counts-per-million normalised before a log(x+1) transformation applied (WGCNA, ESTIMATE) or variance-stabilising-transformation transformed (DESeq2), depending on the requirements of the tool. All bioinformatics analysis was performed on the training set only.

For DE analysis, in order to improve signal to noise, 15 seeds were set, before random splitting of the dataset 80:20. DESeq2 was used on these splits, stratified by the occurrence or non-occurrence of metachronous polyps. For a gene to be considered 'enriched' across folds, it had to be detected in at least 5/15 folds. Of these enriched DEGs, the mean Padj and Log2Fold-Change per DEG were calculated. The ESTIMATE algorithm performs single-sample GSEA to determine the proportion of stromal tissue and immune infiltrates in the cancer, and produces a per-sample score alongside a combined value [2]. WGCNA was performed unsupervised, and modules related to all clinical traits in the data via Pearsons correlation.

Gene set enrichment analysis was performed using GSEApy in Python 3.X using MSigDB gene-sets. Gene networks were visualised using NetworkX via STRING database API. Kaplan Meier survival analysis was performed using Lifelines in Python 3.X, exclusively on the hold-out validation participants

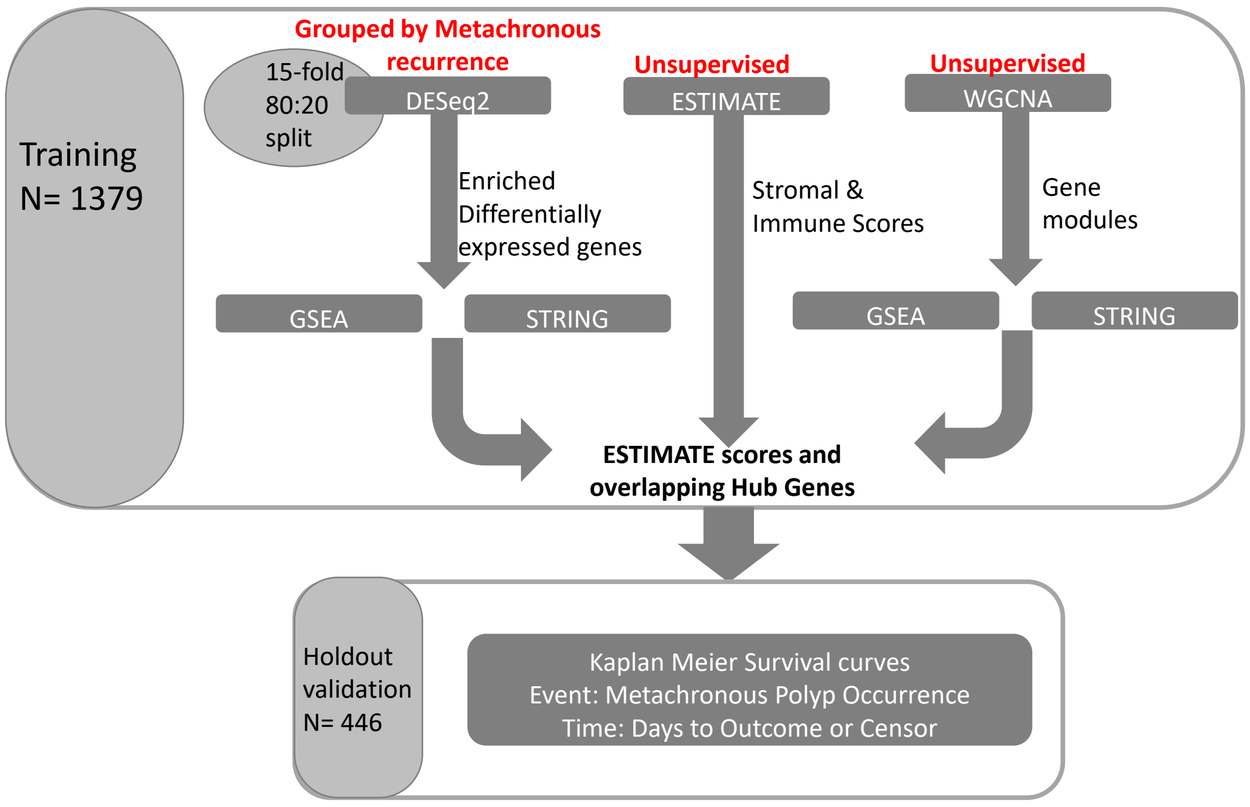


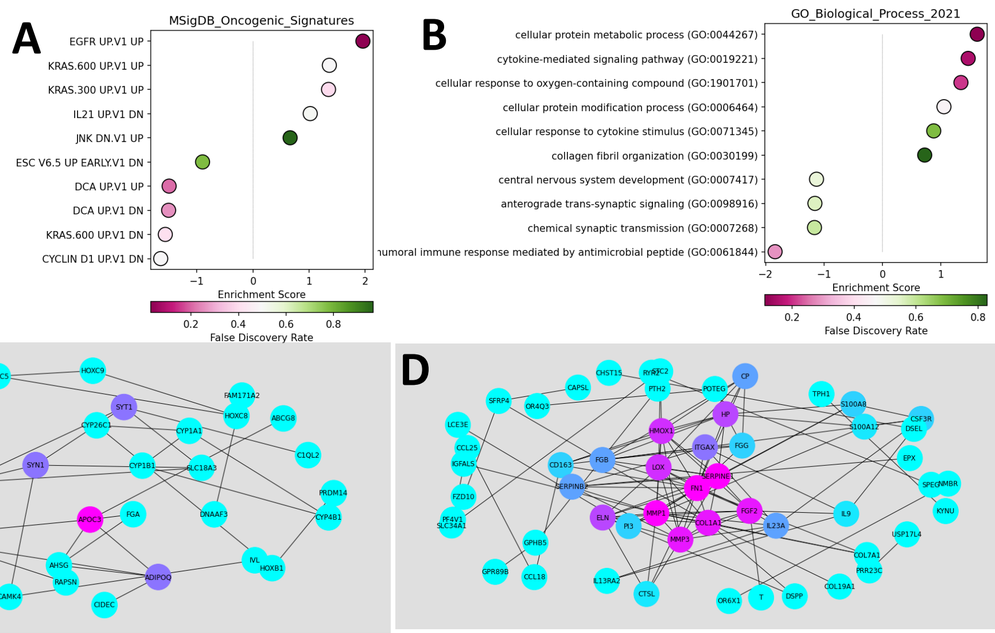
Fig. 1. Experimental design used to create semantic features from INCISE data. DESeq2 returns lists of differentially expressed genes (padj<0.05, Wald stat). ESTIMATE returns a per-patient score of both Stromal and Immune components of a bulk RNA-seq dataset. WGCNA creates a topological overlap matrix containing co-expression modules. An eigengene is formed from this (a dimensionality reduction of the module genes) which is then associated with clinical traits. Both DESeq2 genes and WGCNA can be passed through gene set enrichment analysis and STRING for further inference

Results

Deseq2 Differential Expression Analysis

- GSEA identified trends toward increased EGFR and KRAS signalling in metachronous individuals
- The index polyps of metachronous individuals display greater extracellular invasion and immune system interaction than non-metachronous individuals
- Network analysis of positive DEGs identified several hub genes which may associate with recurrence, each of which has been noted as effecting extracellular matrix formation and maintenance

Fig. 2 A) Differentially expressed genes ranked by L2FC across MSigDB Oncogenic signatures B) Gene Ontology Biological Processes. C) Negative DEGs in Yes-Metachronous events, mapped to STRING, coloured by edge count. D) Positive DEGs

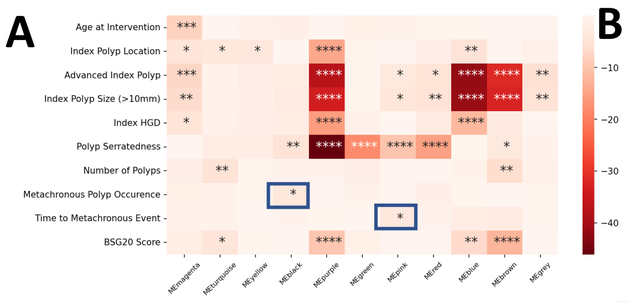


ESTIMATE Scores

- There was no association between index polyp stromal or immune score and polyp recurrence

Weighted Gene Coexpression Network Analysis

- WGCNA identified modules correlated with both metachronous occurrence and time to event
- Six heavily connected genes in the DESeq2 analysis were also identified as hub genes in WGCNA
- The metachronous event module was similarly associated with ECM invasion and immunity as observed above
- The time-to-event module was associated with mitotic activity and histone organisation



B

Gene Name	Entrez Function
HMOX1	Heme catabolism
COL1A1	Collagen formation
CD163	Local inflammation
CTSL	Collagen catabolism
FN1	Cell adhesion & migration
LOX	Collagen crosslinking

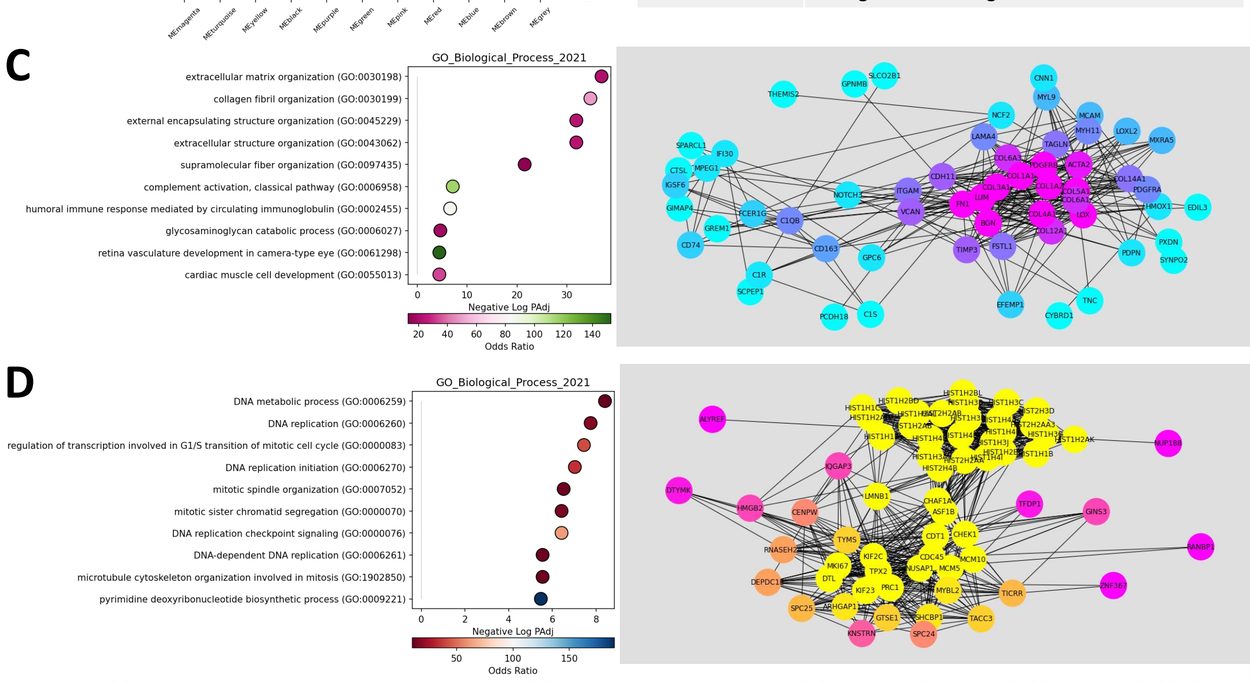


Fig. 3 A) WGCNA module eigengene correlation with INCISE clinical trait data (* p<0.05, ** p<0.01, *** p<0.001, **** p<0.00001, multiple testing corrected). B) Metachronous polyp correlated module hub-genes which overlap with DESeq2 hub genes, with Entez DB function. C) GSEA and STRING network of metachronous correlated module. D) Time-to-event correlated module.

Kaplan Meier Survival on Holdout Validation

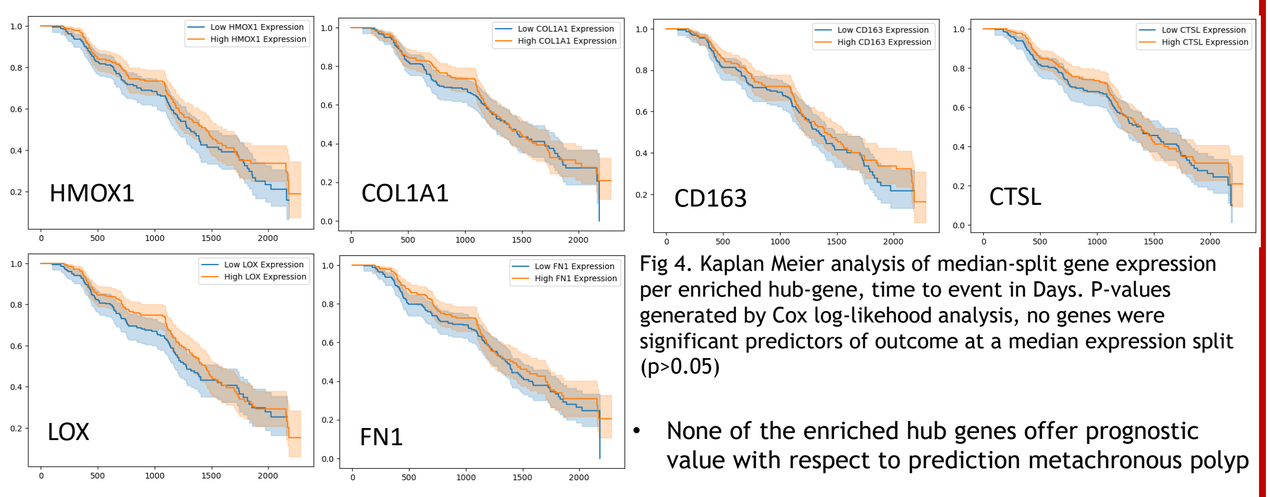


Fig 4. Kaplan Meier analysis of median-split gene expression per enriched hub-gene, time to event in Days. P-values generated by Cox log-likehood analysis, no genes were significant predictors of outcome at a median expression split (p>0.05)

- None of the enriched hub genes offer prognostic value with respect to prediction metachronous polyp

Key Findings and Conclusions

- Index polyps in individuals with metachronous risk may be more likely to invade surrounding tissue and interact with the immune system
- Ultimately the six hub genes identified offered no prognostic information on metachronous event in a hold-out set, suggesting a negligible population-level effect size

References

[1] Johnstone, Mark S., Lynch, Gerard, Park, James, McSorley, Stephen and Edwards, Joanne (2021) Novel methods of risk stratifying patients for metachronous, pre-malignant colorectal polyps: a systematic review. Critical Reviews in Oncology/Haematology, 164, 103421. (doi: 10.1016/j.critrevonc.2021.103421) (PMID:34246774)

[2] Yoshihara, K., Shahmoradgoli, M., Martínez, E. *et al.* Inferring tumour purity and stromal and immune cell admixture from expression data. *Nat Commun* **4**, 2612 (2013). https://doi.org/10.1038/ncomms3612